



Department of Electrical and Electronics Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE

Course Code & Title: EE8552 Power Electronics Renewable Energy Sources

Name of the Faculty member (s): Mr. N. Ganesh

Innovative Practice Description

- **Unit / Topic:** Unit I / Power Semi-conductor Devices
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 2
- **Activity Chosen:** One Minute Paper
- **Justification:**

Commutation circuits are required for Silicon Control Rectifier (SCR). SCR can be turned on by applying appropriate positive gate voltage between the gate and cathode terminals. An SCR doesn't need any gate voltage to stay turned ON. Even if the gate voltage is removed, the SCR doesn't stop conducting. So, the gate has no control flow of current from anode to cathode and the SCR cannot be turned OFF through the gate terminal. The SCR can be brought back to the forward blocking state from the forward conduction state only by reducing the anode or forward current below the holding current level. The turn OFF process of an SCR is called Commutation. There are mainly two types of SCR commutation techniques: Natural Commutation and Forced Commutation. After teaching the various commutation circuits, I had conducted one minute paper activity for making the students to understand the necessity and types of commutation circuits of SCR and also enhance their learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 5 minutes
- **Details of the Implementation:**

After teaching the concept, gave students one or two minutes to think about the topic without writing anything.

Total Strength is 33.

Photographer: One student – Mr. R. Maha Ganapathi Kameswaran (953619105013)

Reporter: Myself

At the end the Class (Last 15 minutes)



- ✓ I asked the students to think about various methods of commutation SCR for 2 minutes.
- ✓ Then I asked them to list the various methods of commutation SCR for another 1 minute.
- ✓ Finally, I randomly asked students about the methods of commutation SCR for 2 minutes.

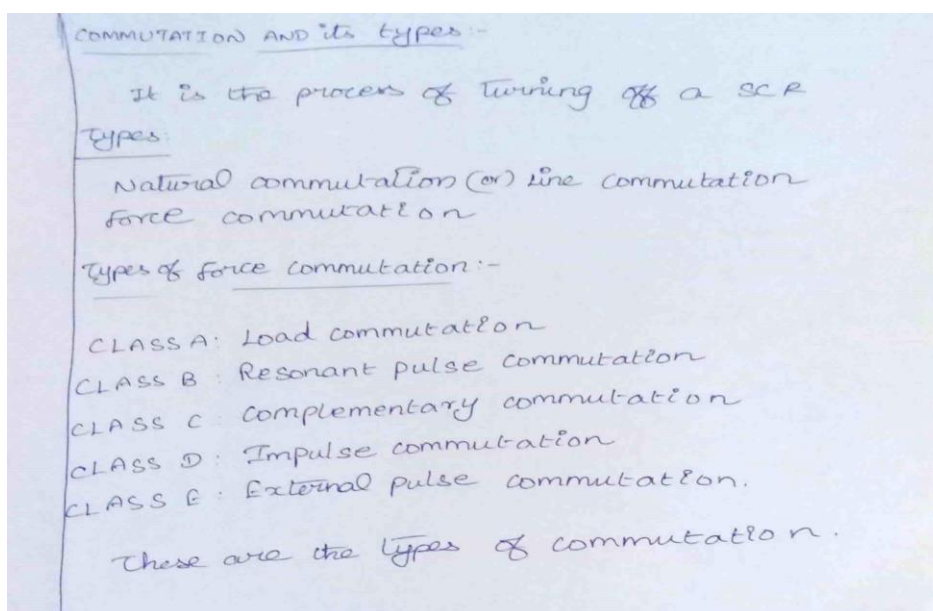
- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 10	PO12	PSO1
CO 1	3	2	1	1	1	1
(1 – Low		2 – Moderate		3 – High)		

- **PO / PSO mapped:**

Innovative practice	PO 7
	1
Justification for correlation	The basic applications of commutation circuits in a sustainable development context were taught to the students, hence it is slightly correlated (level 1).

- **Images / Screenshot of the practice:**





Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students were able to classify the different types of commutation circuits and also they said that they understood the functions of the commutation circuits.

❖ *Benefit of the practice:*

1. Students can able to demonstrate the knowledge of the engineering solution for sustainable development.
2. Students understood the concept which was reflected from their answers for the questions I have asked during discussion session of field visit.
3. The success of the activity was evaluated by asking the same question during IAT – I – **Around 70% of students answered.**
4. Students can able to explain the concepts in detail in the examination.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity, slow learners were not able to answer the question.

References:

- ✓ P.S.Bimbira “Power Electronics” Khanna Publishers, Third Edition, 2003.

Signature of Faculty Member

HOD



Department of Electrical and Electronics Engineering

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Course Code & Title: EE8552 Power Electronics Renewable Energy Sources

Name of the Faculty member (s): Mr. N. Ganesh

Innovative Practice Description

- **Unit / Topic:** Unit IV / Inverters
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 15
- **Activity Chosen:** Mini Map
- **Justification:**

Three Phase Voltage Source Inverters (VSI) is six – step bridge inverter which uses six switches. A step is defined as a change in the firing from one switch to the next switch in the proper sequence. For one cycle of 360° , each step would be of 60° interval for a six – step inverter. The switches would be gated at regular intervals of 60° in proper sequence. There are two possible patterns of gating the switches. In one pattern, each switch conducts for 180° and in the other, each switch conducts for 120° . In both these patterns, gating signals are applied and removed at 60° intervals of the output voltage waveforms. After teaching three phase 180 and 120 Degree mode VSI, I had conducted mini map activity for making the students to understand the operation of three phase 180 and 120 Degree mode VSI and also enhance their learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

After teaching the concept, gave students one or two minutes to think about the topic without writing anything.

Total Strength is 33.

Photographer: One student – Mr. R. Venkateshan (953619105025)

Reporter: Myself

At the end the Class (Last 15 minutes)

- ✓ I asked the students to compare the 180 and 120 Degree conduction mode for 2 minutes.



- ✓ Then I asked them to write the differences between 180 and 120 Degree conduction mode for another 2 minutes.
- ✓ Finally, I randomly asked students about the operation of three phase 180 and 120 Degree mode VSI for 2 minutes.
- ✓

• **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 5	PO10	PO12	PSO1	PSO3
CO 4	3	2	1	2	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 7
	1
Justification for correlation	The basic applications of solar energy in an environmental context are taught to the students, hence it is moderately correlated (level 2).

• **Images / Screenshot of the practice:**

Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

Students were able to classify the different types of commutation circuits and also they said that they understood the functions of the commutation circuits.



❖ ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society
2. Students understood the concept which was reflected from their answers for the questions I have asked.
3. The success of the activity was evaluated by asking the same question during IAT – III – **Around 80% of students answered.**
4. Students can able to explain the concepts in detail in the examination.

❖ ***Challenges faced in implementation:***

1. Time utilization for conducting activity, slow learners were not able to answer the question.

References:

- ✓ P.S.Bimbira “Power Electronics” Khanna Publishers, Third Edition, 2003.

Signature of Faculty Member

HOD